

LINUX LAB Assignment

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1. Print "Hello World"

Problem statement

Write a shell script to print "Hello world" on the terminal.

Script Code :

```
# This script prints Hello world! to the terminal  
echo "Hello, world!"
```

Output :-

Hello, World!

Explanation :- This script uses echo Command, which prints whatever follows it to the terminal screen.

2. Check Even or Odd Number

Problem statement

Write a shell script to check if a given number is even or odd.

Script Code [with comments] :-

```
# Script to check if a given number is even or odd.  
echo "Enter a number"
```

```
read num
```

```
if [  $((num \% 2))$  -eq 0 ]
```

```
then
```

```
    echo "$num is even"
```

```
else
```

```
    echo "$num is odd"
```

```
fi
```

Output ↴

Enter a number

5

5 is odd

Short Explanation ↴ The script reads a number, uses the modulo operator % to check if the remainder when divided by 2 is 0 (even). If not, it is odd.

3. factorial of a Number ↴

Problem statement ↴ write a shell script to calculate the factorial of a given Number.

Script Code :-

```
# Script to calculate factorial of a given number
echo "Enter a number:"
```

```
read n
```

```
fact = 1
```

```
i = 1
```

```
while [ $i -le $n ]
```

```
do
```

```
    fact = $(($fact * i))
```

```
    i = $(($i + 1))
```

```
done
```

```
echo "Factorial of $n is $fact"
```


Expected Output :-

Enter a number :-

5

factorial of 5 is 120

Short Explanation :- The script uses a while loop to multiply numbers from 1 to the Given number, storing the result in fact.

4. Problem statement :- Write a shell script (count-lwc.sh) that takes a filename as input and prints the number of lines, words and character in the file.

Script Code (with Comments)

```
#!/bin/bash
```

```
# count-lwc.sh
```

```
# usage: ./count-lwc.sh filename.txt
```

```
# check if exactly one argument is provided
```

```
if [ $# -ne 1 ]; then
```

```
echo "Usage: $0 <filename>"
```

```
exit 1
```

```
fi
```

```
# check if the specified file exists
```

```
if [ ! -f "$1" ]; then
```

```
echo "file not found."
```

```
exit 1
```

```
fi
```



```
# check if the specified file exists
# Count lines, words and characters using wc
lines = $(wc -l < "$1")
words = $(wc -w < "$1")
chars = $(wc -m < "$1")
```

```
# Display results
echo "Lines: $lines"
echo "Words: $words"
echo "Characters: $chars"
```

Short Explanation of logic used:-

- The script first checks if exactly one argument (the filename) is provided.
- It then verifies if the file exists.
- The script uses the wc Command.
 - wc -l for lines.
 - wc -w for words.
 - wc -m for characters.
- Each value is stored in a variable and printed with descriptive labels.

5. Problem statement - Write a shell script (palindrome.sh) that checks whether a given non-negative integer is a palindrome.

Linux shell script code:-

```
#!/bin/bash
```

```
# palindrome-num.sh
```

```
# Usage: ./palindrome-num.sh 1221
```

```
# check if exactly one argument is provided
```

```
if [ $# -ne 1 ]; then
```

```
    echo "Usage: $0 <non-negative integer>"
```

```
    exit 1
```

```
fi
```

```
n="$1"
```

```
# check if input is non-negative integer
```

```
if ! [[ $n =~ ^[0-9]+$ ]]; then
```

```
    echo "Must be a non-negative integer."
```

```
    exit 1
```

```
fi
```

```
orig="$n"
```

```
rev = 0
```

```
# Reverse the number
```

```
while [ "$n" -gt 0 ]; do
```

```
    rev=$((rev * 10 + (n % 10)))
```

```
    n=$((n / 10))
```

```
done
```

```
# check palindrome
```

```
[ "$orig" -eq "$rev" ] && echo "$orig is palindrome." || echo "$orig  
not a palindrome."
```

Short Explanation ↴

- checks there is exactly one argument and that it non-negative integers.
- stores the original number in orig.
- Uses loop to reverse the digits of input.
- Compares the reversed value to the original; if equal, the number is palindrome, other wise it is not a palindrome number.


```
#!/bin/bash
```

```
# palindrome-num.sh
```

```
# Usage: ./palindrome-num.sh 1221
```

```
if [ $# -ne 1 ] ; then
```

```
echo "Usage: $0 <non-negative integers>"
```

```
exit 1
```

```
fi
```

```
n="$1"
```

```
if ! [[ $n =~ ^[0-9]+$ ]] ; then
```

```
echo "Must be a non-negative integers"
```

```
exit 1
```

```
fi
```

```
orig = "$n"
```

```
rev = 0
```

```
while [ "$n" -gt 0 ] ; do
```

```
rev=$((rev*10 + (n%10)))
```

```
n=$((n/10))
```

```
done
```

```
[ "$orig" -eq "$rev" ] && echo "$orig is a palindrome." || echo "$orig is  
not palindrome."
```

```
#!/bin/bash
# palindrome-num.sh
# Usage: ./palindrome-num.sh 1221
```

```
if [ $# -ne 1 ]; then
    echo "Usage: $0 <non-negative integer>"
    exit 1
fi
n="$1"
if ! [[ $n =~ ^[0-9]+$ ]]; then
    echo "Must be a non-negative integer"
    exit 1
fi
```

```
orig="$n"
rev=0
while [ "$n" -gt 0 ]; do
    rev=$((rev*10 + (n%10)))
    n=$((n/10))
done
```

```
if [ "$orig" -eq "$rev" ] && echo "$orig is a palindrome." || echo "$orig is
not a palindrome"
```